

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First / Second Semester 2024-2025

Comprehensive Examination
(EC-3 Make-up)

Course No. : AIMLCZG565/DSECLZG565
Course Title : Machine Learning
Nature of Exam : Open Book
Weightage : 40%
Duration : 2 Hours
Date of Exam : **** (FN/AN)

No. of Pages	=
No. of Questions	=

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Please Note:

A **maximum of 12 main questions** with Q.Nos. 1 to N.

Q.1. [4 Marks]

A data science team is building a **linear regression model** to predict **house prices** using **120 correlated features** (area, number of rooms, locality index, proximity to schools, etc.). The training dataset has **8,000 samples**, but the validation error fluctuates significantly across folds.

To control overfitting, the team applies **Ridge Regression** and experiments with different values of the regularization parameter λ .

They observe the following behaviors:

- **Case 1:** $\lambda = 0$
- **Case 2:** $\lambda = 0.01$
- **Case 3:** $\lambda = 10^4$

Answer the following questions:

(a) For each case above, describe the expected behavior of the **model coefficients** and the **bias–variance tradeoff**. (3 Marks)

(b) In Case 3, the team notices that all features still have **non-zero coefficients**, although many are extremely small. Explain **why Ridge Regression does not perform feature selection**, even when λ is very large. (1 Mark)

Solution:

(a) Effect of λ on coefficients & bias–variance

Award marks **case-wise** (1 mark per case).

Case 1: $\lambda = 0$ (1 mark)

Expected points (any **two**):

- No regularization $\rightarrow$ equivalent to **ordinary least squares**
- Coefficients can become **large and unstable**

- **Low bias, high variance**
- High risk of **overfitting**, especially with correlated features

Case 2: $\lambda = 0.01$ (1 mark)

Expected points (any **two**):

- Mild penalty applied $\rightarrow$ coefficients **shrunk but meaningful**
- Correlated features share weights more smoothly
- **Balanced bias–variance tradeoff**
- Typically **best generalization** performance

Case 3: $\lambda = 10^4$ (1 mark)

Expected points (any **two**):

- Strong penalty dominates the loss function
- Coefficients shrink **very close to zero**
- **High bias, low variance**
- Model underfits $\rightarrow$ predictions become overly smooth / near-constant

(b) Why Ridge does NOT perform feature selection (1 mark)

Award full marks if **any two** points are mentioned:

- Ridge uses **L2 penalty** ($\lambda \sum \beta_j^2$) $\rightarrow$ smooth shrinkage
- Coefficients are **penalized continuously**, not sparsely
- L2 penalty **never forces coefficients exactly to zero**
- All features remain in the model, only with reduced influence
- Feature selection requires **L1 penalty (Lasso)**

Q.2.

[5 Marks]

As a part of efforts to improve students' performance in the exams, you have been given the data showing number of study hours spent by students, their gender and their final results as pass or fail. Using this sample dataset, apply Naïve Bayes classification technique, to classify the test case {No of study hours = 3.5, Gender="male"} either as "Pass", or "Fail".

No of study hours	Gender	Final result
4.5	Male	Pass
7	Female	Pass
2	Male	Fail
4	Female	Fail
2.5	Male	Fail
3	Female	Fail
8.3	Male	Fail
8	Female	Pass
9	Male	Pass

Solution:

1. Prior Probabilities (1 Mark)

Pass count = 4 Fail count = 5

$P(\text{Pass}) = 4/9 = 0.4444444444$

$P(\text{Fail}) = 5/9 = 0.5555555556$

2. Continuous Feature: Study Hours (Gaussian Naïve Bayes) (1 Mark)

Pass class hours: 4.5, 7, 8, 9

$$\text{Mean } (\mu_{\text{pass}}) = (4.5 + 7 + 8 + 9) / 4 = 7.125$$

$$\text{Variance } (\sigma^2_{\text{pass}}) = [(4.5-7.125)^2 + (7-7.125)^2 + (8-7.125)^2 + (9-7.125)^2] / 4 = 2.796875$$

Fail class hours: 2, 4, 2.5, 3, 8.3

$$\text{Mean } (\mu_{\text{fail}}) = (2 + 4 + 2.5 + 3 + 8.3) / 5 = 3.96$$

$$\text{Variance } (\sigma^2_{\text{fail}}) = [(2-3.96)^2 + (4-3.96)^2 + (2.5-3.96)^2 + (3-3.96)^2 + (8.3-3.96)^2] / 5 = 5.1464$$

3. Posterior Probability Calculation (3 Marks)

Gaussian likelihood formula:

$$P(x|y) = (1 / \sqrt{2\pi\sigma^2}) * \exp(- (x-\mu)^2 / (2\sigma^2))$$

Likelihood for study hours = 3.5

$$P(3.5 | \text{Pass}) = 0.02276907768$$

$$P(3.5 | \text{Fail}) = 0.17227808857$$

Gender likelihoods:

$$P(\text{Male} | \text{Pass}) = 2/4 = 0.5$$

$$P(\text{Male} | \text{Fail}) = 3/5 = 0.6$$

Posterior probabilities (unnormalized):

$$P(\text{Pass} | X) \propto (4/9) \times 0.02276907768 \times 0.5 = 0.00505979504$$

$$P(\text{Fail} | X) \propto (5/9) \times 0.17227808857 \times 0.6 = 0.05742602952$$

Final Classification

Since $P(\text{Fail} | X) > P(\text{Pass} | X)$, the predicted class is: FAIL

Q.3.

[6 Marks]

A consumer electronics company wants to predict the scaled-down selling price (continuous) of a new gadget based on two features: (i) Size (in cm) and (ii) Warranty period (in months). The training set has mixed magnitudes, so you must work in a comparable feature space to compute neighborhoods reliably. For a query gadget Q with Size = 7 cm and Warranty = 18 months, produce two estimates: (1) a standard k-NN regression estimate using $k = 4$ (simple mean of the 4 nearest prices), and (2) a locally weighted linear regression estimate using a Gaussian kernel around the query and ONE gradient-descent (GD) update.

Training data (Size, Warranty, Price):

P0(4,6,6.0), P1(6,12,7.5), P2(7,24,12.0), P3(8,6,8.0), P4(10,12,9.0), P5(12,24,11.0), P6(5,24,10.5), P7(9,18,10.0)

Instructions:

- Apply min-max normalization separately to Size and Warranty using the min/max from the training set.
- Compute Euclidean distances in the normalized 2D space and find the 4 nearest neighbors.
- Standard k-NN regression: unweighted mean of the 4 neighbor prices.
- Locally weighted linear regression: $\hat{y} = w_0 + w_1 x_1 + w_2 x_2$, where (x_1, x_2) are normalized features.
- Kernel: $K(d) = \exp(-d^2 / (2b^2))$, with $b = 2$.
- Start from $w_0 = 1.5$, $w_1 = 0.8$, $w_2 = 0.4$ and learning rate $\alpha = 0.1$.
- Perform exactly ONE GD update using only the 4 nearest neighbors

Answer the following questions:

- (a) Compute the standard 4-NN prediction for Q. [2 marks]
- (b) Compute $K(d)$ for the 4 neighbors, perform ONE GD update, and report $(w0_new, w1_new, w2_new)$ and the locally weighted prediction for Q after the update. [3 marks]
- (c) State one reason why the two predictions differ for this dataset. [1 mark]

Solution

- (a) Standard 4-NN Regression [2 marks]

Normalization ranges (from training set): Size min=4, max=12; Warranty min=6, max=24. (✓ 0.5 mark)

Normalized query: $x1_q = (7-4)/(12-4) = 0.3750$, $x2_q = (18-6)/(24-6) = 0.6667$. (✓ 0.5 mark)

Compute normalized Euclidean distances and select the 4 nearest neighbors. (✓ 0.5 mark)

Nearest 4 neighbors (point, distance, price):

P7: $d=0.2500$, $y=10.00$

P2: $d=0.3333$, $y=12.00$

P1: $d=0.3560$, $y=7.50$

P6: $d=0.4167$, $y=10.50$

Standard 4-NN prediction = $\text{mean}(y \text{ of } 4 \text{ NNs}) = 10.0000$. (✓ 0.5 mark)

- (b) Locally Weighted Linear Regression – 1 GD iteration [3 marks]

Kernel $K(d) = \exp(-d^2/(2b^2))$, $b=2$; start weights $(w0, w1, w2) = (1.5, 0.8, 0.4)$; $\alpha=0.1$. (✓ 0.5 mark)

Compute $K(d)$, $\hat{y}$, error $e = y - \hat{y}$, and weighted terms over the 4-NN set. (✓ 1.0 mark)

Pt	Size	War	x1	x2	Y	d	K(d)	$\hat{y}$	e	$K \cdot e$
P7	9	18	0.6250	0.6667	10.00	0.2500	0.9922	2.2667	7.7333	7.6732
P2	7	24	0.3750	1.0000	12.00	0.3333	0.9862	2.2000	9.8000	9.6648
P1	6	12	0.2500	0.3333	7.50	0.3560	0.9843	1.8333	5.6667	5.5776
P6	5	24	0.1250	1.0000	10.50	0.4167	0.9785	2.0000	8.5000	8.3175

Weighted sums:

$\sum K \cdot e = 31.2331$

$\sum K \cdot e \cdot x1 = 10.8541$

$\sum K \cdot e \cdot x2 = 24.9570$

(✓ 0.75 mark)

Update ($\alpha=0.1$):

$w0_new = 1.5 + 0.1 \times (31.2331) = 4.6233$

$w1_new = 0.8 + 0.1 \times (10.8541) = 1.8854$

$w2_new = 0.4 + 0.1 \times (24.9570) = 2.8957$

(✓ 0.75 mark)

Prediction at Q:

Before GD: $\hat{y} = 2.0667$

After 1 GD iteration: $\hat{y} = 7.2608$

(✓ 0.5 mark)

- (c) Why the two predictions differ (crisp) [1 mark]

Standard k-NN uses an unweighted mean, so all 4 neighbors influence equally. Locally weighted regression uses $K(d)$, so the closest neighbors dominate the update; also it fits a local linear model in (Size, Warranty) near Q rather than outputting a flat average. (✓ 1 mark)

Q.4.

[6 Marks]

A telecom company is building a machine learning system to predict **customer churn (Yes/No)** using historical usage and billing data.

The dataset contains **20,000 customers** and **30 features** (call duration, recharge frequency, complaints, etc.).

The data science team experiments with three ensemble approaches:

- **Bagging with decision trees**
- **Random Forest**
- **AdaBoost with decision stumps**

During model evaluation, the team observes the following:

- Individual decision trees are **high-variance** models.
- Each base classifier trained independently achieves an accuracy of approximately **70%**.
- The business requirement emphasizes **robustness and generalization**, not just single-model accuracy.

Answer the following:

(a) The team trains 100 decision trees using **Bagging** and 100 weak learners using **AdaBoost**. Explain, with reference to the training process, **how each method creates diversity among its base learners**. Your answer must clearly distinguish **data-level randomness** from **error-driven focus**.

[2 marks]

(b) When switching from Bagging to **Random Forest**, the team notices improved validation performance. Identify the **additional source of randomness** introduced by Random Forest and explain **numerically or structurally** why this reduces correlation among trees compared to standard Bagging. [2 marks]

(c) Suppose the final ensemble uses **3 independently trained classifiers**, each with accuracy **0.7**, and predictions are combined using **majority voting**.

1. Compute the **probability that the ensemble prediction is correct**.
2. Based on your calculation, state whether the ensemble is **better than an individual classifier**, and justify numerically. [2 marks]

Solution:

(a) Diversity creation mechanisms (2 marks)

Bagging (Bootstrap Aggregating):

- Each decision tree is trained on a **different bootstrap sample** drawn *with replacement* from the same dataset
- Sampling randomness changes the **training distribution** seen by each tree
- Trees are trained **independently and in parallel**

✓ Diversity source: **data sampling randomness**

AdaBoost:

- Learners are trained **sequentially**
- Misclassified samples receive **higher weights**
- Each new learner focuses on **harder examples** missed earlier

✓ Diversity source: **error-driven reweighting**

Marking:

- Bagging explanation: 1 mark
- AdaBoost explanation + contrast: 1 mark

(b) Extra randomness in Random Forest (*2 marks*)

Additional randomness:

At each split, Random Forest considers only a **random subset of features** (e.g., $\sqrt{30} \approx 5$ features).

Why it helps (structural reasoning):

- In Bagging, strong features (e.g., “number of complaints”) may dominate splits across many trees
- Feature subsampling **forces different trees to use different predictors**
- This **reduces correlation** among trees
- Lower correlation + averaging $\Rightarrow$ **greater variance reduction**

(c) Majority voting calculation (*2 marks*)

Each classifier accuracy: $p=0.7$

Majority voting is correct if **at least 2 out of 3 classifiers** are correct.

$$P(3 \text{ correct}) = 0.7^3 = 0.343$$

$$P(2 \text{ correct}) = (3/2)(0.7)^2(0.3) = 3 \times 0.49 \times 0.3 = 0.441$$

$$P(\text{ensemble correct}) = 0.343 + 0.441 = 0.784$$

Individual classifier accuracy = **70%**

Ensemble accuracy = **78.4%**

✓ **Yes, the ensemble is better**, with a clear numerical improvement of **8.4 percentage points**

Q.5.

[6 Marks]

A reliability team is analyzing machine vibration data collected from four machines, represented by a single standardized vibration feature:

$$X = \{-4, -2, 0, 3\}$$

They believe the data is generated by two latent operating modes and model it using a Gaussian Mixture Model (GMM) with two components.

Initial parameters:

$$\pi_1 = 0.6, \pi_2 = 0.4$$

$$\mu_1 = -3, \mu_2 = 2$$

$$\sigma_1^2 = \sigma_2^2 = 1$$

Gaussian density:

$$N(x | \mu, \sigma^2) = (1 / \sqrt{2\pi\sigma^2}) \exp(-(x-\mu)^2 / (2\sigma^2))$$

(a) Perform the E-step. Compute the responsibilities γ_{i1} and γ_{i2} for each data point. [2 marks]

(b) Using the mixture likelihoods obtained above, compute the log-likelihood of the data at iteration 0. [0.5 mark]

(c) Perform the M-step and compute the updated mixing weights, means, and variances. [2 marks]

(d) Now apply one iteration of K-Means with $k = 2$ using initial centroids -3 and 2 . Compute the updated centroids and compare them with the updated GMM means. State one reason for any difference observed. [1.5 mark]

Solution

a) E-step, responsibilities (rounded): [2 marks]

$$x = -4: \gamma_1 \approx 1.0000, \gamma_2 \approx 0.0000$$

$$x = -2: \gamma_1 \approx 0.9996, \gamma_2 \approx 0.0004$$

$$x = 0: \gamma_1 \approx 0.1096, \gamma_2 \approx 0.8904$$

$$x = 3: \gamma_1 \approx 0.0000, \gamma_2 \approx 1.0000$$

b) Log-likelihood at iteration 0: [0.5 marks]

$$\log p(X) \approx -9.9135$$

c) M-step updates: [2 marks]

$$N_1 = \sum_i \gamma_{i1} \approx 2.10926, \quad N_2 = \sum_i \gamma_{i2} \approx 1.89074$$

$$N_1 \approx 2.1093, N_2 \approx 1.8907 \text{ [0.5 marks]}$$

Updated weights

$$\pi_1^{new} = N_1/4 = 0.52732, \quad \pi_2^{new} = N_2/4 = 0.47268$$

Updated means

$$\mu_1^{new} = \frac{\sum_i \gamma_{i1} x_i}{N_1} \approx \frac{(-4)(1) + (-2)(0.999631) + (0)(0.109629) + (3)(\approx 0)}{2.10926} \approx -2.84425$$

$$\mu_2^{new} = \frac{\sum_i \gamma_{i2} x_i}{N_2} \approx \frac{(-4)(\approx 0) + (-2)(0.000369) + (0)(0.890371) + (3)(1)}{1.89074} \approx 1.58629$$

Updated variances

$$(\sigma_1^2)^{new} = \frac{\sum_i \gamma_{i1} (x_i - \mu_1^{new})^2}{N_1} \approx 1.39154$$

$$(\sigma_2^2)^{new} = \frac{\sum_i \gamma_{i2} (x_i - \mu_2^{new})^2}{N_2} \approx 2.24450$$

$$\pi_{1_new} \approx 0.5273, \pi_{2_new} \approx 0.4727 \text{ [0.5 marks]}$$

$$\mu_{1_new} \approx -2.8443, \mu_{2_new} \approx 1.5863 \text{ [0.5 marks]}$$

$\sigma_{1^2_new} \approx 1.3915, \sigma_{2^2_new} \approx 2.2445$ [0.5 marks]

d) K-Means comparison:

- Cluster 1 (near -3): -4, -2
- Cluster 2 (near 2): 0, 3 (since $|0 - 2| = 2 < |0 + 3| = 3$)

Updated centroids:

$$c_1 = \frac{-4 + (-2)}{2} = -3, \quad c_2 = \frac{0 + 3}{2} = 1.5$$

Assignments: [0.5 marks]

Cluster 1: $\{-4, -2\} \rightarrow \text{centroid} = -3$

Cluster 2: $\{0, 3\} \rightarrow \text{centroid} = 1.5$

The centroids do not match the GMM means [0.5 marks]

Reason: K-Means uses hard assignments and does not model variance, while GMM uses soft responsibilities and explicitly models spread. [0.5 marks]

Q.6.

[3 Marks]

A logistics company wants to predict Delivery Time (in minutes) for short intra-city deliveries using Gradient Boosting Regression with squared loss.

Five training samples are given below:

Job	Traffic Level	Distance (km)	Vehicle	Actual Time (y)
J1	High	3	Bike	34
J2	High	6	Van	52
J3	Low	4	Bike	28
J4	Low	8	Van	46
J5	High	5	Bike	40

The Gradient Boosting setup is as follows:

- Initial model F_0 predicts a constant value equal to the mean of y
- Squared loss is used, so pseudo-residuals are $r_i = y_i - F_0(x_i)$
- Learning rate $\eta = 0.2$

The first weak learner $h_1(x)$ is a regression tree defined as:

- If Traffic Level = Low $\rightarrow$ Leaf L_A
- Else (Traffic Level = High):
 - If Distance $\leq 4 \rightarrow$ Leaf L_B
 - Else $\rightarrow$ Leaf L_C

Each leaf predicts the average residual of the samples reaching that leaf.

(a) Compute F_0 and the pseudo-residuals for all samples. [1 mark]

(b) Compute the leaf predictions L_A , L_B , and L_C . [1 mark]

(c) Compute the updated prediction $F_1(x)$ and the new residual for Job J2 after one boosting iteration. [1 mark]

Solution

a)

$$F_0 = \text{mean}(y) = 40 \text{ (0.25)}$$

Residual of all samples from table below (0.75 marks)

Job	y	F ₀	Residual (y - F ₀)	Traffic	Distance	Leaf
J1	34	40	-6	High	3	L_B
J2	52	40	12	High	6	L_C
J3	28	40	-12	Low	4	L_A
J4	46	40	6	Low	8	L_A
J5	40	40	0	High	5	L_C

b) Leaf predictions based on the above table and from table below (1 mark)

Leaf predictions (average residual per leaf)

Leaf	Jobs	Residuals	Leaf Value
L_A	J3,J4	-12, +6	-3
L_B	J1	-6	-6
L_C	J2,J5	+12, 0	6

c)

Update for Job J2 [1 mark]

Given $\eta=0.2$

For J2: Traffic = High and Distance = 6 (>4)

So, Leaf LC $\rightarrow h_1(J_2)=6$

$$F_1(J_2) = F_0 + \eta h_1(J_2) = 40 + 0.2(6) = 41.2$$

$$r_{J_2 \text{ new}} = y_{J_2} - F_1(J_2) = 52 - 41.2 = 10.8$$

Job	F ₀	Leaf	$h_1(x)$	$F_1 = F_0 + \eta h_1$	New Residual
J2	40	L_C	6	41.2	10.8

Q.7.

[8 Marks]

A financial institution is using a Support Vector Machine (SVM) to classify transactions as High Risk (+1) or Low Risk (-1). Each transaction is represented using two structured numeric features x_1 and x_2 . In later stages, the bank plans to include unstructured text embeddings from transaction remarks.

The initial model is trained using a linear SVM on the following clean, linearly separable dataset.

Training Data

Point	x_1	x_2	Class y
P1	1	4	+1
P2	2	3	+1
P3	4	1	-1
P4	5	2	-1

Assume that in the optimal maximum-margin solution, only P2 and P3 are support vectors to answer the questions below.

(a) Using geometric reasoning, identify the equation of the maximum-margin separating hyperplane for the given data. Clearly justify your choice based on the support vectors. [2 marks]

(b) Assuming that the support vectors satisfy the canonical margin conditions $y_i(w^T x_i + b) = 1$, compute a valid weight vector w and bias b . Write the resulting decision function $f(x) = w^T x + b$. [2 marks]

(c) State the condition on the Lagrange multipliers α_i that identifies a support vector in the dual formulation of SVM. Explain why only P2 and P3 influence the final classifier. [1.5 marks]

(d) Compute $\|w\|$ and the margin width $2/\|w\|$. What does a larger margin imply about the classifier's robustness? [2 marks]

(e) To incorporate text embeddings, the bank switches to a kernelized SVM. Consider the kernels:

$$K_1(x, z) = (x^T z + 2)^2$$

$$K_2(x, z) = \exp(-0.5\|x - z\|^2)$$

$$K_3(x, z) = \sin(x^T z)$$

State one reason why kernelized SVMs may become computationally expensive. [0.5 mark]

Solution :

(a) The closest opposite-class points are P2 (2,3) and P3 (4,1). The decision boundary lies midway between them and is perpendicular to the line joining them. The midpoint is (3,2) and the boundary equation is $x_2 = x_1 - 1$.

[1 mark for identifying SVs and midpoint, 1 mark for correct equation]

(b) From $x_2 = x_1 - 1$, we obtain $x_1 - x_2 - 1 = 0$, hence a normal vector $w = [1, -1]^T$ and $b = -1$. Rescaling to canonical form gives $w = [0.5, -0.5]^T$ and $b = -0.5$. The decision function is $f(x) = 0.5x_1 - 0.5x_2 - 0.5$.

[1 mark for direction, 1 mark for valid canonical parameters]

(c) In the dual formulation, a point is a support vector if $\alpha_i > 0$. [0.5]

Only P2 and P3 have non-zero α_i , so they alone contribute to $w = \sum \alpha_i y_i x_i$ and define the classifier. [1 mark]

(d) $\|w\| = \sqrt{(0.5)^2 + (-0.5)^2} = \sqrt{0.5} \approx 0.707$. The margin width is $2/\|w\| \approx 2.83$. A larger margin implies greater robustness to noise and better generalization.

[1 mark for computation, 1 mark for interpretation]

(e) Kernelized SVMs can be computationally expensive due to the need to compute and store large kernel matrices. [0.5 mark]

Q.8.

[2 Marks]

A bank trains two Decision Tree models to predict loan default:

Model A:

- The tree is allowed to grow without depth restriction.
- A leaf node may contain even a single training sample.
- Training Accuracy = 99%
- Test Accuracy = 71%

Model B:

- The tree depth is restricted to 4 levels.
- Each leaf must contain at least 25 training samples.
- Training Accuracy = 86%
- Test Accuracy = 83%

(a) Which model is overfitting? [0.5 mark]

(b) Why does the other model perform well? [0.5 mark]

(c) Briefly explain how the stopping conditions influence model complexity and generalization. [1 mark]

Solution:

(a) Model A is overfitting because it has very high training accuracy but much lower test accuracy. [0.5 mark]

(b) Model B restricts complexity, which improves test performance. [0.5 mark]

(c) Allowing unlimited depth and very small leaf size makes the tree highly complex, enabling it to fit noise in the training data. This increases variance and reduces generalization. [1 mark]
